RiskPulse AI

Real-Time Financial Risk Intelligence, Tactical Index Rebalancing & Portfolio Stress Testing

Hackathon: S&P; Global & CRISIL Campus Hackathon 2026

Candidate Name: [REPLACE WITH YOUR FULL NAME]

College / Campus: [REPLACE WITH YOUR COLLEGE NAME]

College Email ID: [REPLACE WITH YOUR COLLEGE EMAIL]

Value Proposition: Bridging unstructured financial narrative signals to deterministic quantitative index rebalancing and Basel III portfolio stress testing.

Problem Statement & Strategic Approach

Overcoming Information Asymmetry and Disconnected Risk Workflows

Core Industry Challenges:	The RiskPulse AI Solution:
• **Unstructured Overload:** Critical signals lurk in earnings calls, SEC subpoenas, and FinTwit noise without structured quantitative quantification. • **Lagging Transmission:** Traditional portfolio risk tools rely on trailing historical price correlations rather than forward-looking narrative signals. • **Disconnected Modules:** Market risk and banking credit risk operate in isolated silos, lacking explainable unified risk triggers.	• **Central AI/NLP Risk Engine:** Normalizes text into structured signals: Sentiment (-1 to +1), 20 Event Classes, and 1-10 Severity Impact. • **Downstream Module A (Tactical Rebalancing):** De-weights constituents facing high-impact adverse events subject to turnover & box constraints. • **Downstream Module B (Stress Testing):** Translates event signals into Basel III Expected Credit Loss (ECL) & duration shocks on a $500M bank portfolio.

System Architecture & Data Flow

Modular, Zero-Key Resilient Full-Stack Architecture

Component	Technology	Key Responsibilities
Data Ingestion	Python / HTTPX / Adapters	NewsAPI (optional), FinTwit simulation, 240+ deterministic synthetic records.
Preprocessing & Entities	Regex / Dict Mappers	Cashtag ($AAPL) resolution, 15 constituent aliases, HTML cleansing, deduplication.
AI / NLP Risk Engine	FinBERT + Lexicon Fallback	Continuous sentiment (-1.0 to +1.0), 20 event taxonomies, 1-10 impact scoring.
Storage & DB	SQLAlchemy / SQLite / Postgres	Normalized risk signals, rebalance runs, portfolio holdings, immutable audit trails.
Module A: Rebalancer	NumPy / Constrained Optimizer	Exponential weighting, [2%-18%] box constraints, 25% turnover limit, 50bps deadband.
Module B: Stress Engine	Financial Valuation Library	$500M bank portfolio, Basel III Expected Credit Loss (ECL), duration price sensitivity.
Frontend UI	React / Vite / Tailwind / Recharts	Institutional light-theme analytics platform with 10 dedicated operational views.

AI / NLP Risk Engine & Signal Explainability

Transforming Unstructured Text into Actionable Structured Signals

Core Signal Dimensions:

- **Sentiment Score:** Continuous -1.0 to +1.0 with Positive, Negative, Neutral, or Mixed categorical labels.
- **Event Taxonomy:** 20 specialized classes (Corporate Earnings, Regulatory Action, Cyber Incident, Bankruptcy, Interest Rate Hike, etc.).
- **Impact Severity (1 to 10):** Independent from sentiment; synthesizes event baseline, scope (market vs single-stock), and stress indicators.
- **Confidence & Provenance:** Transparent heuristic confidence score and attribution evidence tokens.

Structured Signal Schema Example:

```
{ 'primary_ticker': 'NVDA', 'event_type': 'Corporate Earnings', 'sentiment_score': +0.72 (Positive), 'impact_score': 6/10 (Moderate), 'evidence': ['beat', 'record profit'], 'rationale': 'Revenue expanded 18% beating Wall Street consensus, raising annual guidance.' }
```

Downstream Risk Applications (Modules A & B)

Connecting Sentiment and Event Signals to Capital Decisions

Module A: Tactical Stock Index Rebalancing	Module B: Strategic Portfolio Stress Testing
• **Basket:** 15 core publicly traded US equities across 7 sectors. • **Algorithm:** w_i = w_base * exp(alpha*Sent - beta*Risk). • **Downside Penalty:** Adverse high-impact signals trigger aggressive de-weighting. • **Execution Guardrails:** Iterative projection algorithm enforces min (2%) & max (18%) weights while bounding turnover (≤25%) and eliminating churn with 50bps deadband.	• **Portfolio:** $500M Wholesale Banking Book (Corporate Loans, Sovereign/Corp Bonds, Equities, Derivatives, Cash). • **Basel III ECL:** ECL = PD * LGD * EAD. • **Duration Sensitivity:** Fixed-income valuation shifts: ΔP = -Duration * ΔYield * P. • **Asset Differentiation:** Equities take direct market shocks, corporate debt reflects spread widening, cash remains par.

Empirical Results & Domain Impact

Validated Calculations, Backtest Trajectory & Enterprise Utility

Automated Test Suite Results (13/13 Passed):

- **NLP Engine:** Accurate positive/negative detection, verified negation reversal ('failed to beat' -> adverse, 'did not decline' -> bullish).
- **Rebalancing Optimization:** Guaranteed budget normalization (sum=100.0%), strict box compliance (2% to 18%), turnover strictly bounded.
- **Stress Calculations:** Zero cash impairment, accurate ECL loan provisioning, and verified duration bond price haircuts.

Domain & Business Impact:

- **Institutional Value:** Transforms qualitative research notes into instant risk attribution for portfolio managers and risk committees.
- **Proactive Hedging:** Rebalances indices defensively before quarterly earnings declines are fully priced into benchmark correlations.
- **Regulatory Auditability:** Every action logged with immutable audit IDs and transparent mathematical justifications.

Limitations & Roadmap

Ethical Safeguards, Current Constraints & Future Enhancements

Current Limitations & Safeguards:

- **Synthetic Demonstration Scope:** Prototype utilizes synthetic customer transactions and news records to protect client confidentiality.
- **Simplified Derivative MtM:** Derivatives are modeled through simplified delta/duration exposure approximations rather than full Monte Carlo option surfaces.
- **No Financial Advice:** Platform is an educational risk intelligence prototype, not financial advice or credit rating guarantees.

Roadmap & Future Enhancements:

- **Multi-Lingual NLP:** Expansion to European and Asian financial news languages via multi-lingual transformer models.
- **Non-Linear Derivatives:** Integration of Black-Scholes and Heston volatility surface modeling for structured derivative books.
- **Automated Order Routing:** FIX protocol adapter integration for automated paper-trading execution with institutional brokers.